



Phytic acid/graphene oxide nanocomposites modified electrode for electrochemical sensing of dopamine

Donglei Wang, Fei Xu, Jiajie Hu, Meng Lin *

Key Laboratory for Colloid and Interface Chemistry of State Education Ministry, School of Chemistry and Chemical Engineering, Shandong University, Jinan 250100, China

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ABSTRACT

An electrochemical sensor for determining dopamine was developed by modifying phytic acid/graphene oxide (PA/GO) nanocomposites onto a glassy carbon electrode (GCE). PA functionalized GO was prepared by an ultra-sonication method. Subsequently, the PA/GO nanocomposites were drop-casted on a glassy carbon substrate. The structural feature of the PA/GO modified GCE was confirmed by attenuated total reflection infrared (ATR-IR) spectroscopy. The proposed electrochemical sensor was applied to detect various concentrations of DA by differential pulse voltammetry (DPV). The PA/GO/GCE was considered to be highly sensitive to DA in the range of 0.05–10 μM . In addition, the PA/GO/GCE demonstrated high electrochemical selectivity toward DA in the presence of ascorbic acid (AA) and uric acid (UA). The prepared electrochemical DA sensor was applied for detection of DA in dopamine hydrochloride injection and spiked samples of human urine with satisfactory results.

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1. Introduction

Dopamine (DA) is considered as one of the key neurotransmitters in the brain that implicated in a number of human pathologies [1]. Abnormal levels of DA may lead to neurological disorders, such as Parkinson's disease and schizophrenia [2]. Hence, an accurate and sensitive determination of the concentration of DA has received considerable attention. Due to its high sensitivity and simplicity, electrochemical methods are widely used for the determination of biological and environmental samples, such as DA, folic acid and glutathione et al. [3–5]. During the electro-analysis of DA, it is extremely important to avoid interferences from the co-existing compounds in the biological samples, such as ascorbic acid (AA) and uric acid (UA), whose oxidation potentials are close to that of DA [6,7]. In order to address the problems, different kinds of modified electrodes such as conducting polymer nanocomposite [7], metal oxide nanomaterial [8] and carbon material [9] modified electrodes have been developed to improve the selectivity and the sensitivity toward DA detection.

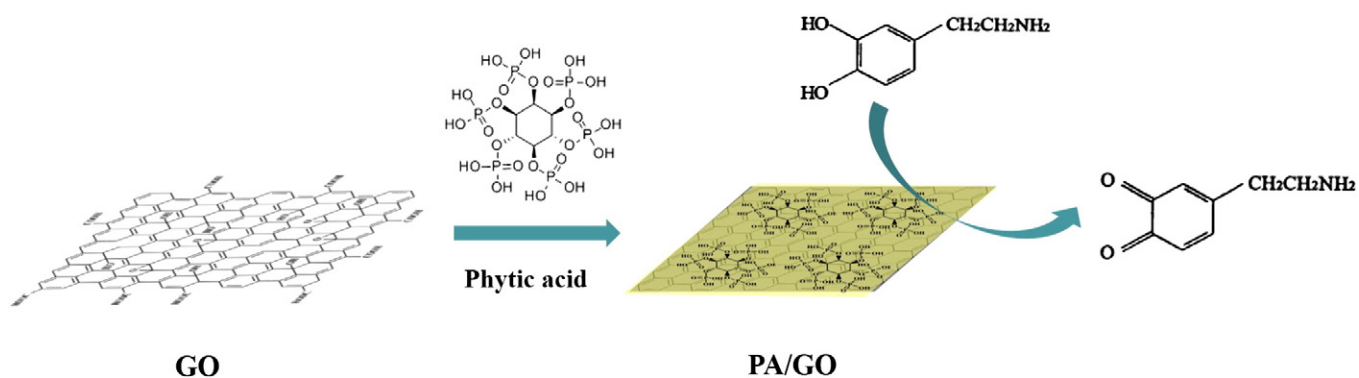
Phytic acid (PA) is a naturally derived material from legume seeds, cereal grains, and beans [10]. Resulting from the unique properties, such as nontoxic, biocompatible, and green to the environment, PA has been attracted increasing interest for food additive, water treatment agent and corrosion prevention of metal [9]. Moreover, PA has six

phosphates groups attached symmetrically to a cyclohexanehexol ring which is liable to interact with positively charged metal ions or biomolecules [11,12]. For example, Yang and co-workers used a phytate functionalized multi-walled carbon nanotubes modified electrode for determining trace concentration of copper ions [13]. Additionally, PA can attach onto the surface of carbon materials for forming a network structure through hydrogen-bond interaction [13]. Due to the larger available surface area, the good electrical performances of the PA functionalized carbon materials, and the strong combination between PA to DA by electrostatic interaction, a phytic acid/single-walled carbon nanotube modified electrode has been employed for detection of DA [14].

Up to now, a few kinds of electrochemical DA sensor based on the PA have been developed [14–16]. But to the best of our knowledge, there is still no study on phytic acid/graphene oxide (PA/GO) nanocomposites for DA detection. In this paper, an electrochemical DA sensor based on a PA/GO nanocomposites modified glassy carbon electrode was prepared (Scheme 1). Electrochemical behaviors of the PA/GO/GCE were investigated using differential pulse voltammetry (DPV), and the PA/GO/GCE proved its capability to determine the concentration of DA in the range of 0.05–10 μM and the detection limit of the DA oxidation current was determined to be 16 nM. Moreover, the electrochemical behavior of DA in the presence of AA and UA on PA/GO/GCE was studied. The results showed that the oxidative potentials of DA, AA, and UA can be isolated by the modified electrode, indicating the PA/GO/GCE showed great selectivity toward DA. The modified electrode was also

* Corresponding author.

E-mail address: milin@sdu.edu.cn (M. Lin).



Scheme 1. A schematic of the PA/GO/GCE functionalized electrochemical sensor for DA detection.

applied to the determination of DA in dopamine hydrochloride injection and human urine samples.

2. Experimental

2.1. Materials

Natural graphite powder was purchased from Alfa Aesar. Phytic acid (PA), dopamine (DA), uric acid (UA) and ascorbic acid (AA) were obtained from Sigma-Aldrich. Disodium hydrogen phosphate (Na_2HPO_4) and sodium dihydrogen phosphate (NaH_2PO_4) were obtained from Sinopharm Chemical Reagent (Shanghai, China). The reagents were of analytical grade, and were used without further purification. All solutions were freshly prepared using ultrapure water ($>18 \text{ M}\Omega$, Pall Corp., USA). 10 mM phosphates buffered solution (PBS, 0.5 M NaCl, pH 7.0) prepared by mixing appropriate amounts of Na_2HPO_4 and NaH_2PO_4 was employed as a supporting electrolyte during the electrochemical analysis. Stock solution of DA (5 mM) was prepared in 0.10 M PBS (pH 7.0) and diluted to give various concentrations of DA.

2.2. Fabrication of the modified electrode

Glassy carbon electrode (GCE, diameter: 5 mm) was firstly polished with 1.0 μm , 0.3 μm and 0.05 μm alumina slurry until a mirror surface was established. After washed ultrasonically with ethanol and ultrapure water, respectively, the GCE was cleaned by cycling the potential between -0.3 and $+1.5 \text{ V}$ (vs. SCE) at 100 mV/s in a 0.5 M H_2SO_4 solution until a steady-state electrochemical oxidation and reduction wave was obtained.

Graphene oxide (GO) was synthesized from natural graphite powder using a modified Hummer's method [17]. Prior to the modification processes, 5 mL GO dispersion (1 mg/mL) and 5 mL 0.01 mol/L PA solution were mixed together and further treated with ultra-sonication for

1 h to form a stable PA/GO nanocomposites suspension. Then, 10 μL of the PA/GO nanocomposites dispersion was drop-casted onto the surface of GCE, and dried under an infrared lamp. The obtained PA/GO/GCE was rinsed with water and stored in PBS (pH 7.0).

2.3. Apparatus

All electrochemical measurements were performed on a CHI760C workstation (Chenhua Instruments, China) with a conventional three-electrode system. The modified electrode (GCE) was used as the working electrode, a platinum plate and a saturated calomel electrode (SCE, 3 M KCl) served as the counter electrode and the reference electrode, respectively. Differential pulse voltammetry (DPV) was carried out using a step increment of 5 mV, amplitude of 50 mV, and pulse period of 0.2 s. The morphology of the prepared composites was characterized by scanning electron microscope (SEM, HITACHI-S4700). Attenuated total reflection infrared (ATR-IR) spectra were obtained by infrared spectrophotometry (Bruker, Tensor 27, Bruker Optik, Germany) with 4 cm^{-1} resolution and 100 scans, using an ATR platform by pressing the solid electrodes onto the diamond crystal.

3. Results and discussion

According to the previous report, phytic acid with six phosphate groups can easily interact with the COOH or OH groups of the GO [17]. In order to investigate the formation of the nanocomposites modified GCE, the SEM-analysis and ATR-IR spectrum of GO and PA/GO films are observed. As shown in Fig. 1, PA/GO film (Fig. 1b) exerted little effect on the GO structure (Fig. 1a). With modification of the PA to the surface of GO, the surface shows little departure from the original one presumably due to the small size of the PA molecules. The ATR-IR spectra of GO (red line) and PA/GO (black line) films are observed in Fig. 2. ATR-IR spectra of the GO displays a very wide and strong intensity of vibration

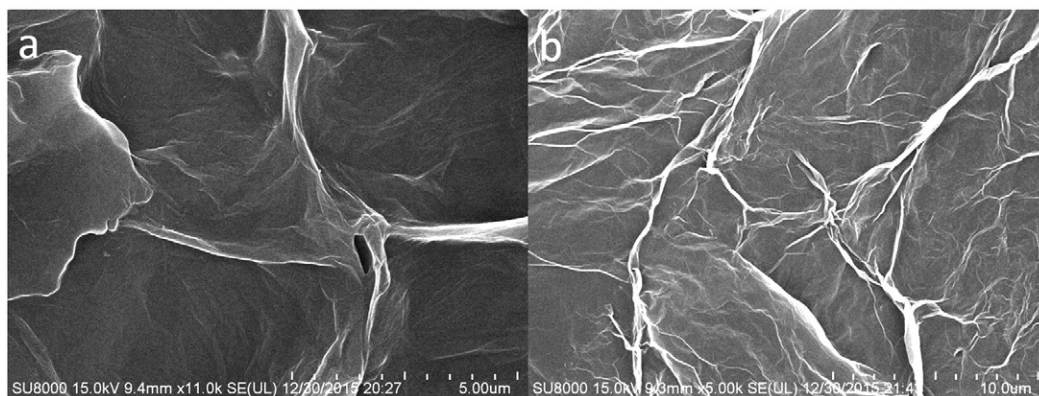


Fig. 1. SEM images of GO (a) and PA/GO (b) films on GCE.

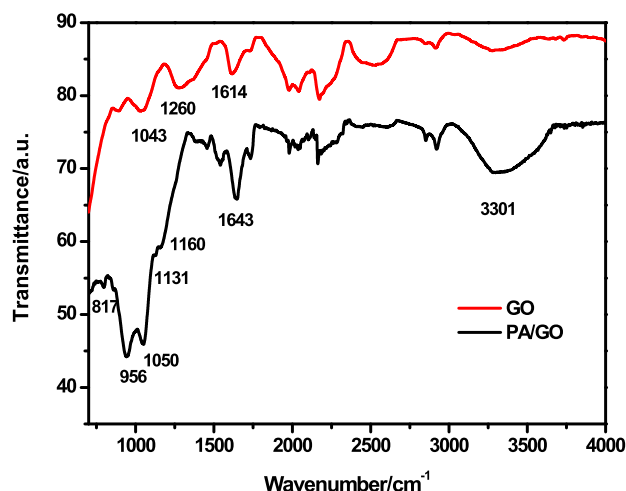


Fig. 2. ATR-IR spectra for the GO (red line) and PA/GO (black line) film modified GCE.

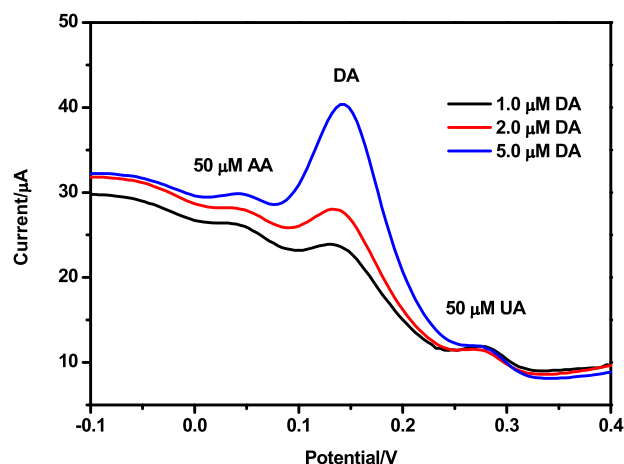


Fig. 4. DPV of the PA/GO/GCE in PBS (10 mM, pH 7.0) containing 5 μM DA, 50 μM AA and 50 μM UA.

peaks of O—H groups at 3301 cm^{-1} , and the C=C stretching, and C—O stretching in C—OH and C—O—C functional groups are analyzed from the peaks located at 1614 , 1260 and 1043 cm^{-1} , respectively [18,19]. Comparing with the absorption peaks of the GO film, the characteristic peaks of the PA/GO are centered at 1050 cm^{-1} and 1643 cm^{-1} , assigning to phosphate radical (PO_4^{3-}) and phosphate hydrogen radical (HPO_4^{2-}), respectively [20]. The peaks are emerged at 956 cm^{-1} and 817 cm^{-1} , corresponding to the vibration of P—O—C bonds, and the peaks at 1131 cm^{-1} and 1160 cm^{-1} are assigned to stretching vibration of P=O bonds [21]. The increased intensity of O—H at 3301 cm^{-1} is attributed to the hydroxyl of PA molecules. The result is in good agreement with the predicted structure of GO functionalized with PA by an ultrasonic mixing method.

In order to study the sensitivity and selectivity of three different types of electrodes (GCE, GO/GCE and PA/GO/GCE) for DA detection, we compared the DPV responses of $5\text{ }\mu\text{M}$ DA in the presence of $50\text{ }\mu\text{M}$ AA and $50\text{ }\mu\text{M}$ UA in 10 mM PBS (pH 7.0). As shown in Fig. 3, the oxidation peaks of AA and DA show only one broad peak around 0.130 V at the GCE and GO/GCE, indicating the two species cannot be distinguished from each other. However, due to the π – π interaction between the GO basal plane and the phenyl structure of DA molecules, an increase of the peak current for DA is observed on GO/GCE. In the case of the PA/GO/GCE, it is obviously seen that the AA peak potential can be resolved from DA. Compared to the low selectivity of the GCE and GO/GCE, the

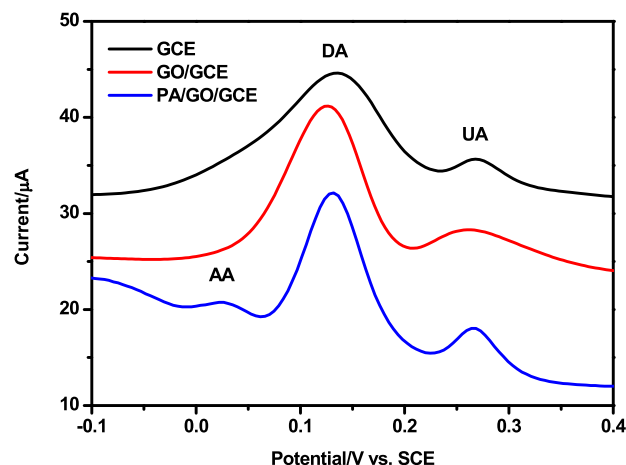


Fig. 3. DPV curves at GCE, GO/GCE and PA/GO/GCE in 10 mM (pH 7.0) PBS containing ternary mixture of $50\text{ }\mu\text{M}$ AA, $5\text{ }\mu\text{M}$ DA and $50\text{ }\mu\text{M}$ UA.

DPV peaks are well-resolved at PA/GO/GCE with the peak potentials at about 0.025 V , 0.130 V and 0.266 V for AA, DA and UA, respectively. The good selectivity of the PA/GO/GCE toward DA is attributed to the surface charge and functional groups of PA/GO nanocomposites, the strong combination between PA to DA by electrostatic interaction could provide an excellent selective interface for the detection of DA [22]. As illustrated in Fig. 4, a series of DPV curves of DA was measured

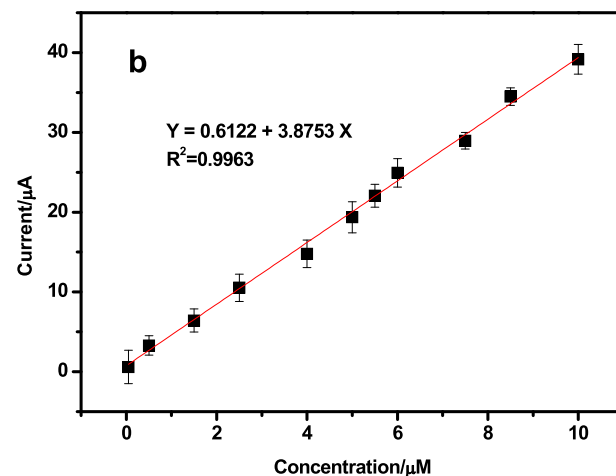
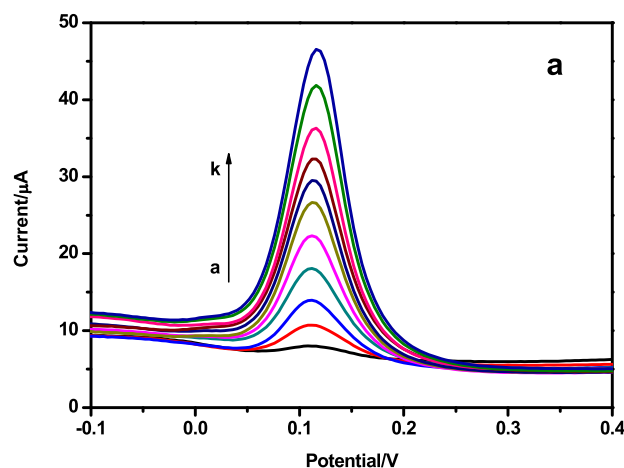


Fig. 5. DPV response of PA/GO nanocomposites modified GCE in the presence of different concentration addition of DA from 0.05 to $10\text{ }\mu\text{M}$ (a–k) in PBS (10 mM , pH 7.0).

Table 1

A comparison of the different methods reported for the detection of DA.

Electrode	Method	Detection limit (μM)	Linear range (μM)	Refs.
PA/SWCNT/Pt	DPV	0.08	0.2–10	[14]
Laccase/SiO ₂ -PA/GCE	Amperometry	0.26	0.99–103.1	[16]
PPy/Fe ₃ O ₄ /rGO/GCE	DPV	0.00233	0.007–1.2	[26]
ZnO/Carbon paste	SWV	0.056	0.3–100	[27]
ZnO/Cu ₂ O/PPy/GCE	DPV	0.04	0.1–130	[28]
Molybdenum(VI) complex/carbon paste	DPV	0.076	0.1–800	[29]
PA/GO/GCE	DPV	0.016	0.05–10	This work

by increasing the concentration of DA while concentrations of AA (50 μM) and UA (50 μM) were kept at a constant value. It can be seen that 50 μM additions of AA and UA did not affect the response to current and peak potential of DA. The results prove that the PA/GO nanocomposites modified electrode exhibits selectively determining DA in the case of other two species [23].

Differential pulse voltammetry (DPV) is a sensitive technique which has been often applied for determination of DA [24,25]. Fig. 5a exhibits DPV responses of PA/GO/GCE toward the addition of DA. Resulting from the oxidation of DA to dopamine-o-quinone, a well-defined current response is observed at 0.130 V [25]. In addition, the peak current of DA increases linearly with the increasing the addition of different concentrations of DA into the electrolyte solution. The calibration curve (Fig. 5b) shows a linear increase in current density with the DA concentration from 0.05 to 10 μM ($R^2 = 0.9963$), the detection limit was calculated as 16 nM ($S/N = 3$), and the sensitivity was $4.58 \mu\text{A } \mu\text{M}^{-1} \text{ cm}^{-2}$, illustrating a high sensitivity of the PA/GO/GCE modified electrode to DA. Numbers of reports for the electrochemical detection of DA are compared with the present work, as listed in Table 1. It is found that the developed electrochemical sensor had a wide linear range and low detection limit, which may be attributed to the synergistic effect of PA and GO nanocomposites. Compared with the electrochemical DA sensors based on metal or metal oxide nanostructures modified electrode, our sensor may not show a lower limit of detection or wider detection range. But, through the contrast test by the representative diagnostic detection of DA in the hospital, all of the results are acceptable. Moreover, our strategy represents a simple one-pot preparation of PA/GO hybrids for DA sensing with a biocompatible, nontoxic and environment friendly natural material.

Five different PA/GO nanocomposites modified electrodes were investigated by DPV measurement of 5 μM DA in 10 mM PBS (pH 7.0). The relative standard deviation (RSD) of peak current is calculated as 4.7%. After testing, the modified electrode was rinsed with PBS (pH 7.0) and stored at 4 $^{\circ}\text{C}$, and the electrochemical response of 5 μM DA lost only 5.9% of the initial response after two weeks. These results indicate that the modified electrode has good reproducibility and stability.

To illustrate the feasibility of the PA/GO/GCE, the modified electrode was implemented to analyze DA in dopamine hydrochloride injection (10 mg/mL, 2 mL per injection) and urine samples by DPV. Prior to the detection, the dopamine hydrochloride injection was diluted into

various concentrations using PBS (10 mM, pH 7.0), and then the developed sensor was immersed into an electrochemical cells which containing the different diluted samples for the DA determination. After centrifuged, filtered through 0.22 μm syringe filter and diluted 100 times with PBS before the measurement, the urine samples were spiked with different amounts of known concentrations of DA for electrochemical analysis. As indicated in Table 2, the values of determination are very close to the declared values, and the recovery is found to be in the range of 97–104.6%. The results suggest that the PA/GO/GCE is reliable and sensitive for the detection of DA.

4. Conclusion

In this paper, determination of DA in the presence of AA and UA was achieved by using a PA/GO nanocomposites modified GCE. Due to the synergistic effect of the PA/GO nanocomposites, the fabricated sensor exhibited high electro-catalytic activities toward the detection of DA, and the results confirmed the suitability of the PA/GO/GCE as an effective sensing strategy for selective DA determination in the presence of AA and UA. The electrochemical performance of the PA/GO/GCE was investigated by DPV technique, and the results revealed that the sensor has highly sensitivity to detect DA in the range of 0.05–10 μM , and the sensor also displayed good reproducibility and stability. Moreover, the availability of the PA/GO nanocomposites can be successfully used for the detection of DA in dopamine hydrochloride injection and human urine samples.

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Table 2Determination of DA levels in different samples using the PA/GO/GCE ($n = 3$).

Sample	DA/ μM added	DA/ μM found	Recovery (%)
Injection	1	1.01	101
	3	2.96	98.67
	5	4.95	99
Urine	1	0.97	97
	3	3.08	102.7
	5	5.23	104.6